

CS 111

Foundations of Program Design
Week 4, Monday Lecture

This lecture

So far, we have used conditional logic (`if / else`) to make one-off decisions in programs where all code runs once, in order from top to bottom. In this lecture, we'll combine conditionals with looping statements to write more complex programs.

FizzBuzz is a program that prints "Fizz" for numbers divisible by 3, "Buzz" for numbers divisible by 5, and "FizzBuzz" for numbers divisible by 15.

For any integer i , its remainder divided by three ($i \% 3$) is either 0, 1, or 2. When $i \% 3 = 0$, we say that i is divisible by 3. Here's an incomplete FizzBuzz program:

```
public class FizzBuzz {  
    public static void main(String[] args) {  
        int i = 0;  
        while (i <= 15) {  
            if (i % 3 == 0) {  
                System.out.println("Fizz");  
            }  
            else {  
                System.out.println(i);  
            }  
            i = i + 1;  
        }  
    }  
}
```

Additional resources

These slides adapt parts 2-1, 2-3, and 2-4:

<https://java-programming.mooc.fi/part-2>